Computer Science

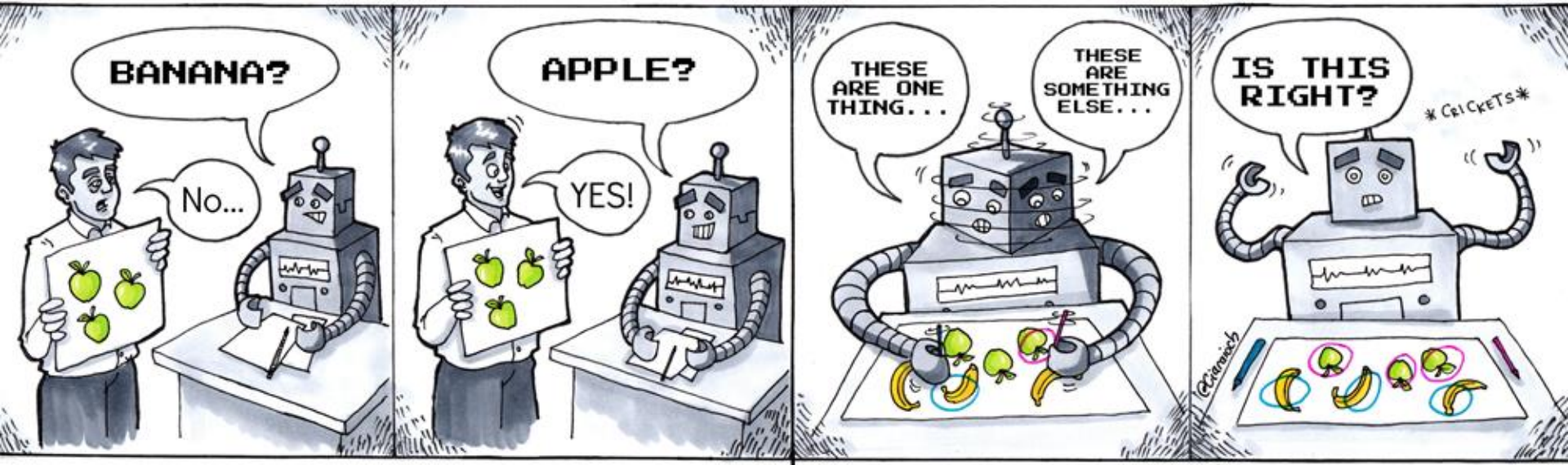
CSC380: Principles of Data Science

Nonlinear Models 2

Xinchen Yu

Outline

- K-Means
- Neural Networks



Supervised Learning

Unsupervised Learning

Task 1 : Group These Set of Document into 3 Groups based on meaning

Doc1 : Health , Medicine, Doctor

Doc 2 : Machine Learning, Computer

Doc 3 : Environment, Planet

Doc 4 : Pollution, Climate Crisis

Doc 5 : Covid, Health, Doctor

Task 1 : Group These Set of Document into 3 Groups based on meaning

Doc 3 : Environment, Planet

Doc1 : Health , Medicine, Doctor

Doc 4 : Pollution, Climate Crisis

Doc 5 : Covid, Health , Doctor

Doc 2 : Machine Learning, Computer

Task 2: Topic modeling

- Provides a summary of a corpus.
- Collected n tweets containing the keyword "bullying", "bullied", etc.
- Extracts k topics: each topic is a list of words with importance weights.
  - A set of words that co-occurs frequently throughout.

"feelings"

"suicide"

"family"

"school"

"verbal bullying"

"physical bullying"

Learning from bullying traces in social media.
Jun-Ming Xu, Kwang-Sung Jun, Xiaojin Zhu, and Amy Bellmore.
In the Conference of North American Chapter of the Association for Computational Linguistics: Human Language Technologies (**NAACL HLT**), 2012. [pdf]

What is unsupervised learning?

unsupervised learning: learning with unlabeled data

What can we expect to learn?

- **Clustering**: obtain partition of the data that are well-separated. A preliminary classification without predefined class labels.
  - As a summary of the data
  - **Exploratory data analysis**: what are the **patterns** even without labels?

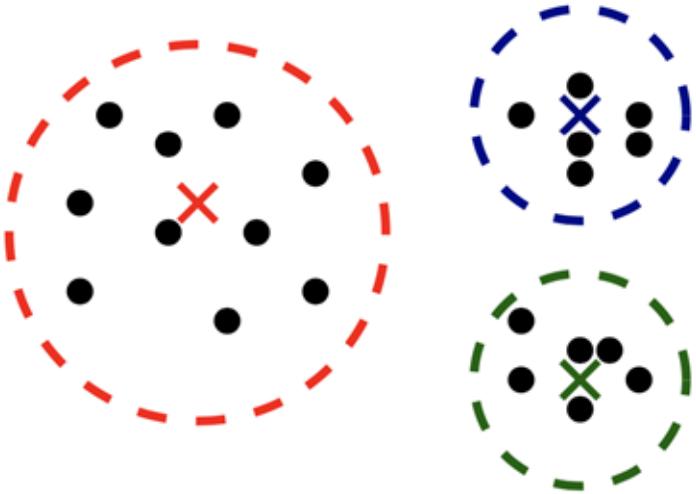
- **Components**: extract common components, e.g., topic modeling given a set of articles: extract the topics that appear frequently.

Clustering

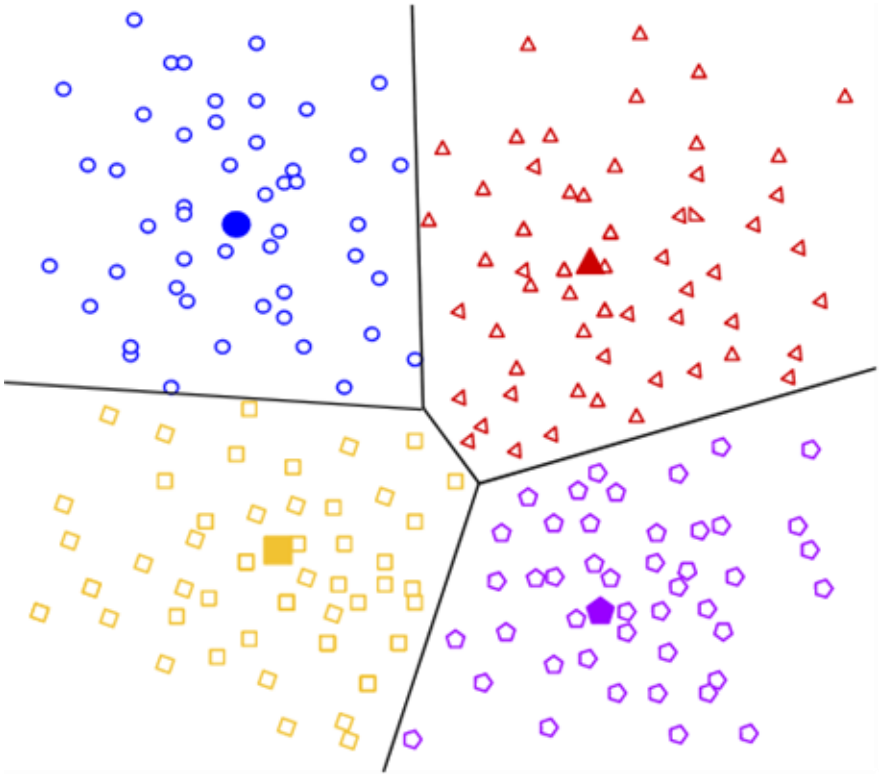
- Input: k: the number of clusters (hyperparameter)

$$S = \{x_1, \dots, x_n\}$$

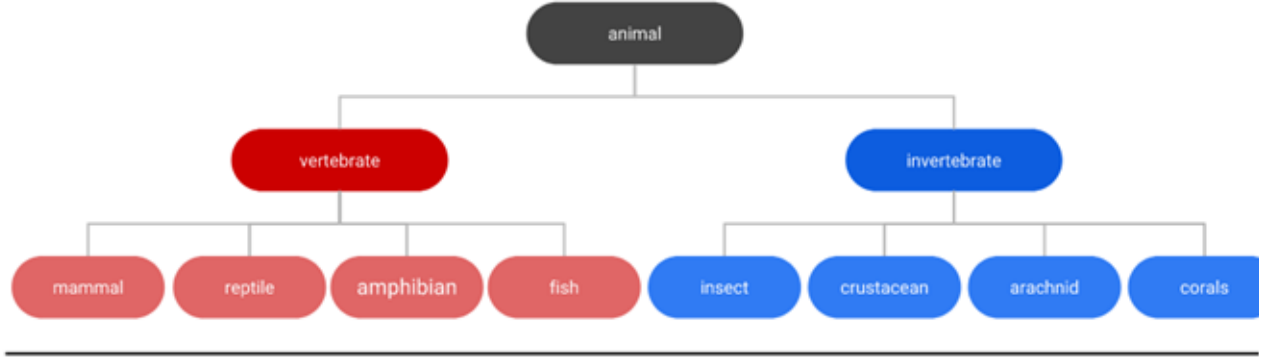
- Output
  - partition $\{G_i\}_{i=1}^{k}$ s.t. $S = \cup_i G_i$ (disjoint union).
  - often, we also obtain 'centroids'

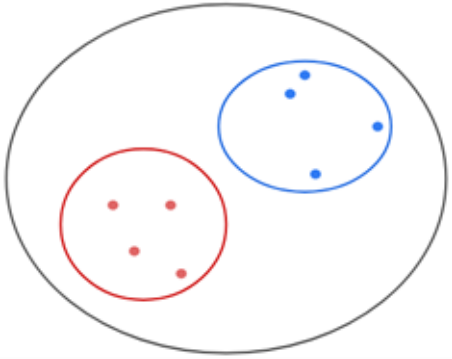


Centroid-based Clustering

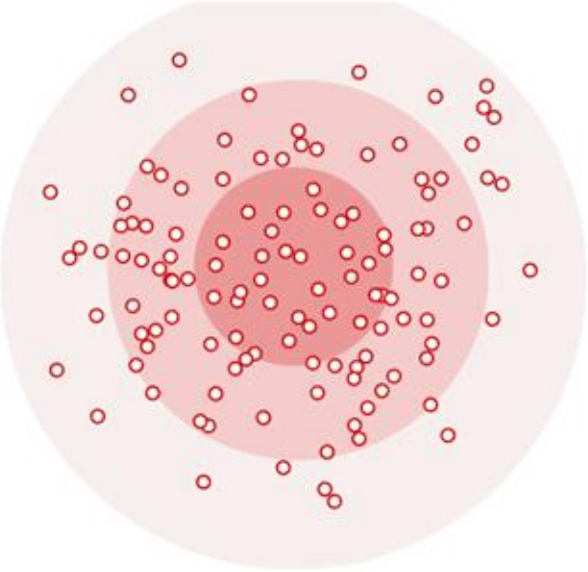


Hierarchical Clustering

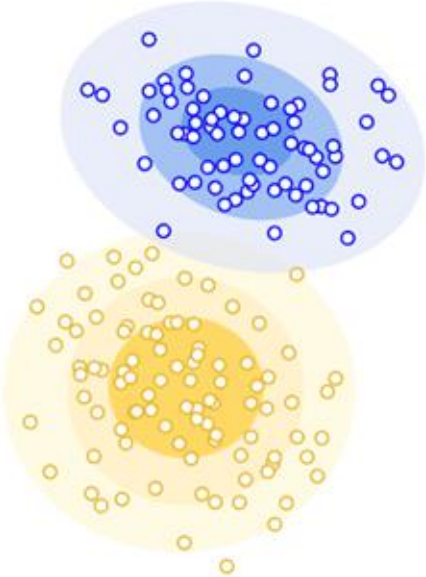




Distribution-based Clustering

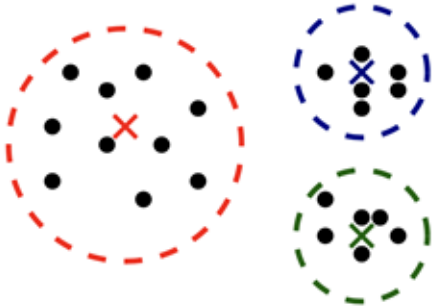


(probabilistic treatment)



k-means: how to define centroids

<u>Lloyd's algorithm</u>: solve it approximately (heuristic)

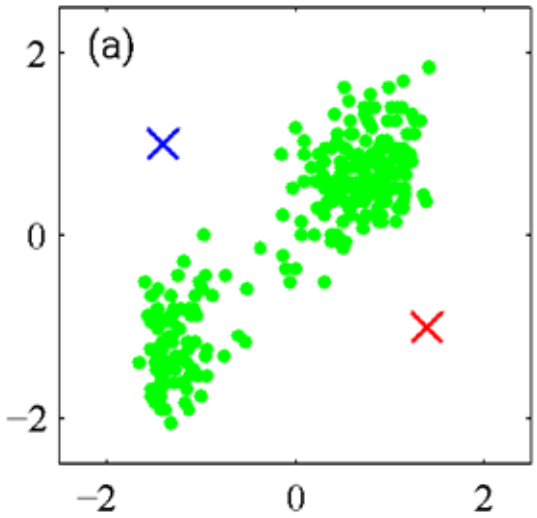


<u>Observation</u>: The chicken-and-egg problem.

- If you knew the cluster assignments… just find the centroids as the average
- If you knew the centroids… make cluster assignments by the closest centroid.

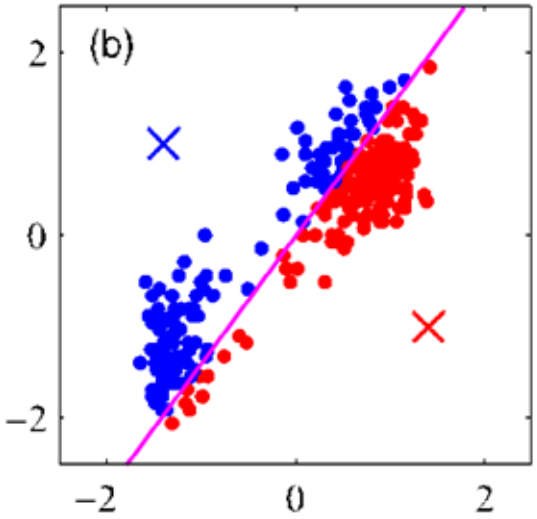
Why not: start from some centroids and then alternate between the two?

Initialization

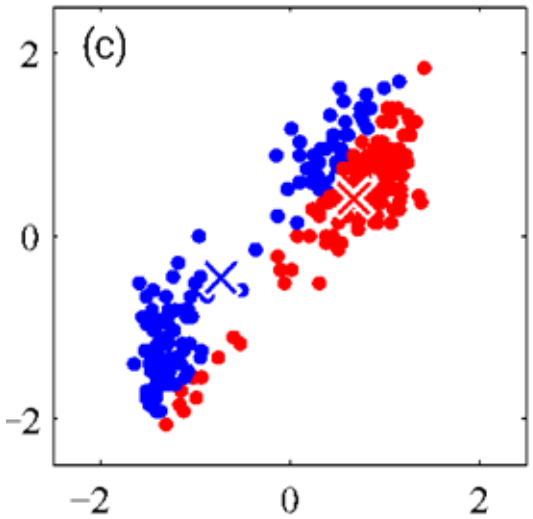


Arbitrary/random initialization of c_1 and c_2

Iteration 1

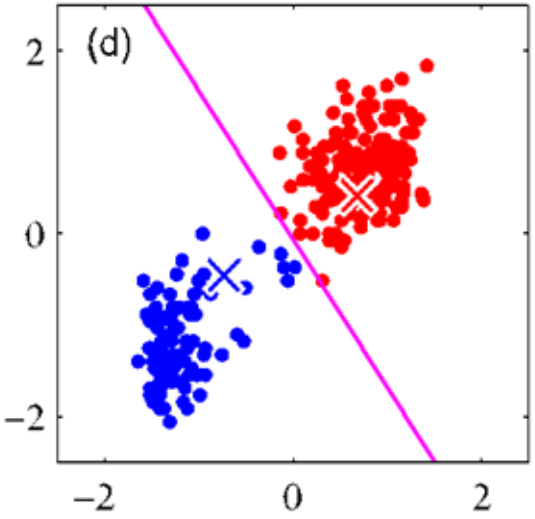


(A) update the cluster assignments.

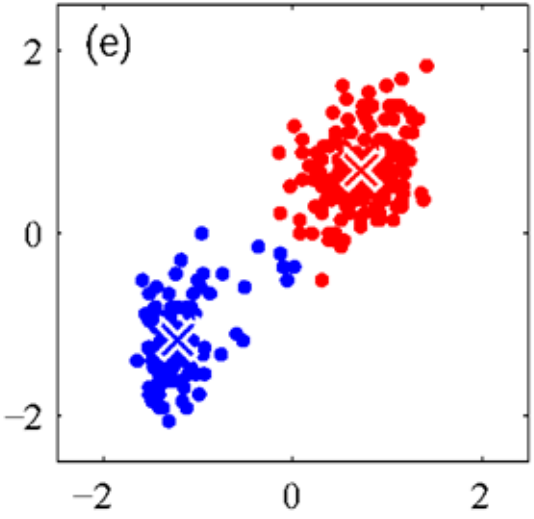


(B) Update the centroids $\{c_j\}$

Iteration 2

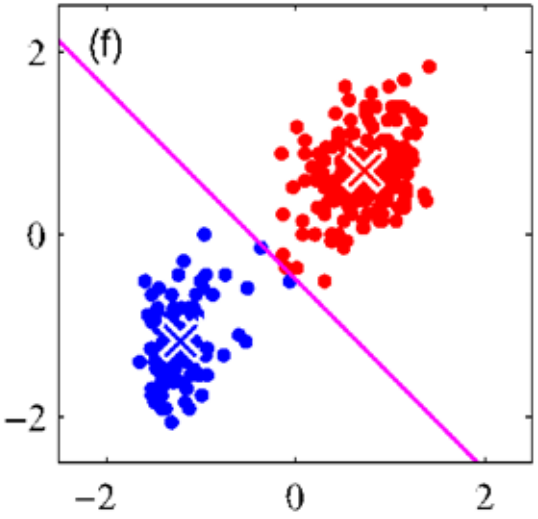


(A) update the cluster assignments.

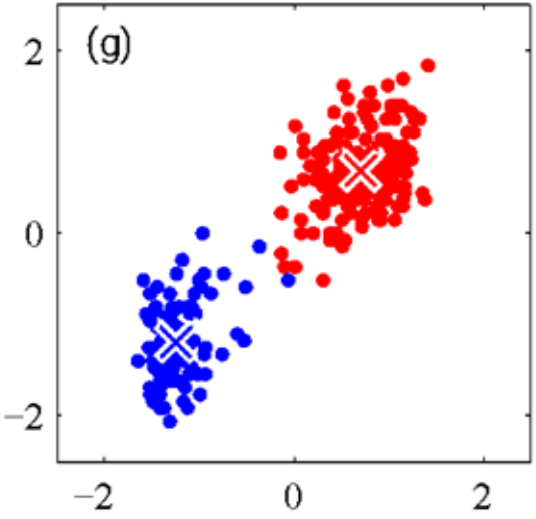


(B) Update the centroids $\{c_j\}$

Iteration 3

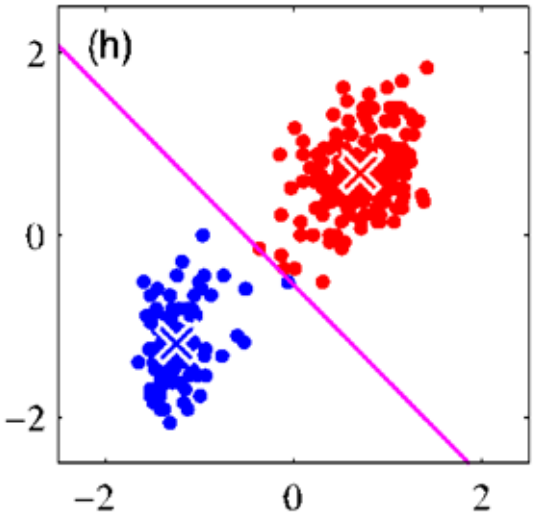


(A) update the cluster assignments.

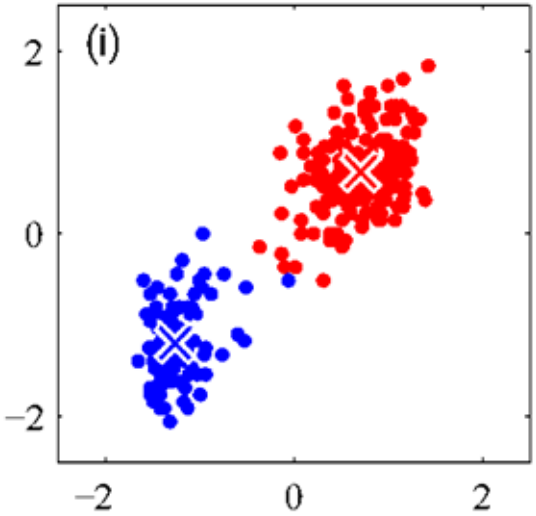


(B) Update the centroids $\{c_j\}$

Iteration 4

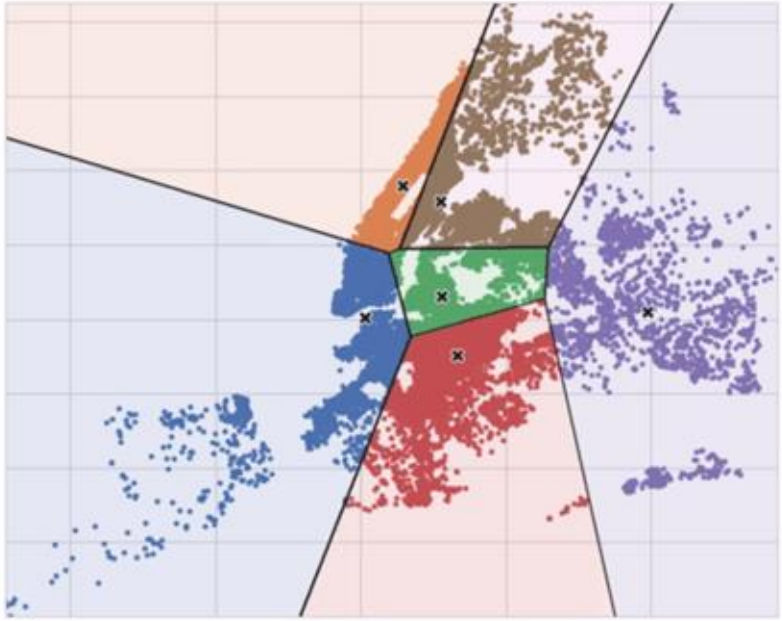


(A) update the cluster assignments.



(B) Update the centroids $\{c_j\}$

Iterating until Convergence



Animation from [Kaggle]

k-means clustering

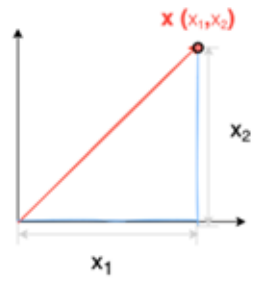
<u>Input</u>: k: num. of clusters, $S = \{x_1, \dots, x_n\}$

[Initialize] Pick $c_1, \dots, c_k$ as randomly selected points from S (see next slides for alternatives)

For t=1,2,…,max_iter

- ***[Assignments]*** $\forall x \in S, \quad a_t(x) = \arg\min_{j \in [k]} \|x - c_j\|_2^2$
- If t $\neq 1$ AND $a_t(x) = a_{t-1}(x), \forall x \in S$
  - break

$$||x||_2 = \sqrt{x_1^2 + x_2^2}$$



k-means clustering

Input: k: num. of clusters, $S = \{x_1, \dots, x_n\}$

[Initialize] Pick $c_1, \dots, c_k$ as randomly selected points from S (see next slides for alternatives)

For t=1,2,…,max_iter

- ***[Assignments]*** $\forall x \in S, \quad a_t(x) = \arg\min_{j \in [k]} \|x - c_j\|_2^2$
- If $t \neq 1$ AND $a_t(x) = a_{t-1}(x), \forall x \in S$
  - break
- ***[Centroids]*** $\forall j \in [k], \quad c_j \leftarrow \text{average}(\{x \in S : a_t(x) = j\})$

Output: $c_1, \dots, c_k$ and $\{a_t(x_i)\}_{i \in [n]}$